

Functional Programming -

1] Lambda

2] map

3] filter

4] reduce

1] Lambda function -

① Lambda is a anonymous
function - nameless
function.

② It is defined using lambda
keyword.

Normal function

```
def fun_name():
```

```
    return sum
```

```
result = fun_name()
```



③ lambda function is a simple throwaway function that we are not going to use in the future.

Characteristics of lambda function

- ① Anonymous - nameless
- ② Single expression - one line
- ③ return value - automatically returns value.

Syntax :-

lambda arguments : expression

10 20

eg. lambda x, y : $x + y$

Program 1 :- create a simple lambda function that adds two

number

Normal function:

```
def summation(x, y):  
    result = x + y  
    return result
```

```
result1 = summation(3, 5)
```

```
print(result1) → 8
```

lambda function :-

Syntax :-

lambda arguments : expression

8 ← add = lambda x, y : (x + y) ^{3 + 5 = 8}

```
print (add (3, 5)) → 8
```

Program 2: check the no. is even

or not

Normal fun:-

```
def is_even(x):
```

```
    if (x % 2 == 0):
```

```
        print("True")
```

is_even(6) → True

$$\begin{array}{r} 3 \\ 2 \overline{)6} \\ \underline{6} \\ 0 \end{array}$$

0 == 0 ✓
6 % 2 == 0

lambda function

is_even = lambda x: x % 2 == 0

print(is_even(6)) → True

Program 3:- Square of no.

Square = lambda x: x ** 2

print(square(5))

o/p - 25

$$5^2 = 25$$

Program 45 Concatenation of String
using lambda.

```
concat = lambda str1, str2 : str1 + str2  
print (concat ("hello", "world"))
```

23 map function – map function in python is useful function that applies a given function to each item of iterable (list, tuple, set, range) and creates a new iterable with result

Syntax :-

```
map (function, iterable)
```

function – normal fun, lambda fun

iterable – list, tuple, set, range

function - def func(x):

function `def square(x):`
`return x ** 2`

iterable `list` `[1, 2, 3, 4, 5]`

1 4 9 16 25 $\leftarrow$ result
 (map object)

`list(map obj)`

o/p - `[1, 4, 9, 16, 25]`

→ when we apply map function
 it return we get map
 object that we convert
 into iterable such as
 list, set, tuple, etc.

Program:- Square of numbers using
 map function.

`numbers = [1, 2, 3, 4, 5]` $\leftarrow$ iterable

`def square(x):` $\leftarrow$ function

return x ** 2

sq_numbers = map(Square, numbers)

↑ 1, 4, 9, 16, 25
map-object ↑ x

print(sq_numbers) → <map-object>

print(list(sq_numbers)) → [1, 4, 9, 16, 25]